A variational characterization of geodesic scattering on simple disks

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Abstract

We give necessary and sufficient conditions for a boundary map to be the geodesic scattering map of a smooth simple Riemannian metric on a fixed disk with a prescribed boundary metric. The conditions depend only on the proposed map and that boundary metric. Boundary invariant functions first determine a normalized conformal boundary coordinate. In the resulting Euclidean disk, length inequalities define a nonempty convex set of densities whose strong L^2 closure has a unique element of least norm. Realizability is equivalent to smooth positivity and the prescribed boundary value of this element, explicit geodesic and Jacobi equation tests, and a scalar area equality. The proof transfers the length inequalities through the closure by integrating over the geodesic family. Strict monotonicity of the area functional then forces equality of boundary distances and hence of scattering maps. The characterization is exact and qualitative; no finite decision procedure or stability estimate is asserted.

1 Introduction

The scattering map records the point and direction at which a geodesic leaves a manifold after entering at its boundary. On a simple surface, every inward direction has a finite flight and distinct boundary endpoints. For a fixed boundary metric, the tangential components of the incoming and outgoing unit covectors give a common space on which to compare these maps. The range question asks which proposed maps are produced by smooth simple Riemannian metrics. It is a consistency question for the directions of rays, before one attempts to recover an interior metric.

This question differs from uniqueness for data already known to come from a metric. Pestov and Uhlmann prove that two-dimensional compact simple Riemannian manifolds are boundary distance rigid, and their Theorem 1.3 recovers the Dirichlet-to-Neumann map from actual scattering data [5]. The metric-to-scattering range question is posed separately in [4, Section 15.1, problem 2]. Bohr and Paternain's Theorem 1.5 characterizes non-Abelian scattering data on an already fixed simple surface [2]; the geometric scattering relation of that surface is an input there. Lens-map realization results for Finsler metrics [1] also concern a different class of interior geometries. The problem considered here allows an arbitrary proposed boundary map and requires a Riemannian realization. The geometric and transport results used below are those developed in [4].

Fix a smooth compact disk M and a smooth positive boundary metric h. The requested characterization is in terms of A and h alone, with no additional interior metric among the data. We give such a characterization by selecting a density before testing realizability.

Main result, informal form. *A map on the entire open unit cotangent bundle of the boundary is realizable if and only if it passes the boundary action and boundary Fourier tests below, and the uniquely selected least-norm density is smooth and positive with the prescribed boundary value, passes*

the fixed geodesic and Jacobi equation tests, and has the area determined by the boundary action. There is no smallness, genericity, or local perturbation assumption.

The boundary Fourier test extracts the traces of holomorphic functions for actual scattering data, and a three-point normalization selects a circle coordinate. One then considers all positive smooth densities on a fixed Euclidean disk whose lengths dominate the proposed boundary distances. This convex set is always nonempty after the boundary tests, even for data that have no realization. Its strong L^2 closure therefore selects one density class without making a metric-existence assumption. The remaining conditions test this fixed class rather than search among simple metrics.

The main analytical difficulty is that restriction to an individual curve is not continuous in L^2 on the disk. Santaló's formula instead controls the error integrated over all geodesics of the selected smooth metric. This yields a pointwise lower bound on its boundary distance, by continuity. A symmetric integration-by-parts identity shows that the action area is strictly increasing under a strict increase of boundary distance with positive mixed derivative. Thus one scalar area equality forces exact equality of all boundary distances. First variation then identifies the scattering map.

The result is a qualitative characterization of the full range. Its tests include infinitely many curves, smoothness of an L^2 minimizer, and exact ODE and scalar conditions. It does not supply a finite computation or quantitative stability, and it does not address non-simple surfaces or Finsler realization.

Section 2 states the conditions and theorem. Section 3 proves the boundary and conformal facts for actual simple metrics. Section 4 proves the universal convex construction and the two directions of the characterization. Throughout, the exit covector is recorded without reversal, Fourier averages have the factor $1/(2\pi)$, and all L^2 spaces and closures of densities are real and use Euclidean area.

2 Boundary tests and the selected density

Choose an orientation, an h-arclength origin, and three distinct positively cyclically ordered boundary points s_1, s_2, s_3. Write $L > 0$ for the h-length of the boundary and identify it with $\mathbb{R}/L\mathbb{Z}$. The positive unit tangent T is chosen so that (T, ν) is positive when ν is the inward unit normal. Then

$$Z = (\mathbb{R}/L\mathbb{Z}) \times (-1, 1)$$

identifies with $B_h^*\partial M$: p denotes the coefficient of the tangential covector $p\,ds$. Start with an arbitrary set map $A : Z \to Z$. The following conditions are evaluated in order. If a condition fails, the map is rejected and later constructions are not needed.

2.1 The boundary action: condition (C1)

Require that A be a smooth diffeomorphism. Write $A(s,p) = (t,q)$ and require the endpoint map

$$E : Z \longrightarrow \mathcal{C}, \qquad E(s,p) = (s,t), \qquad \mathcal{C} = \{(s,t) \in (\mathbb{R}/L\mathbb{Z})^2 : s \neq t\},$$

to be a smooth diffeomorphism. Its inverse defines functions $P(s,t) = p$ and $Q(s,t) = q$. For the unique forward lift $s < t < s + L$, put

$$D(s,t) = \int_s^t Q(s,u)\,du. \tag{1}$$

The integral converges at its lower endpoint since $|Q| < 1$. Its profile $d(s,\delta) = D(s, s+\delta)$, initially $0 < \delta < L$, must extend smoothly to $(\mathbb{R}/L\mathbb{Z}) \times [0, L]$ and satisfy

$$d(s,0) = d(s,L) = 0, \qquad d_\delta(s,0) = 1, \qquad d_\delta(s,L) = -1, \tag{2}$$

$$D > 0, \qquad D(s,t) = D(t,s), \qquad D_s = -P, \qquad D_t = Q, \qquad D_{st} > 0 \quad \text{on } \mathcal{C}. \tag{3}$$

Derivatives of D treat its endpoints as independent variables; hence $D_s = d_s - d_\delta$ and $D_{st} = d_{s\delta} - d_{\delta\delta}$. Define the action area

$$\mathcal{V}(D) = \frac{1}{2\pi} \int_{\mathcal{C}} D D_{st}\, ds\, dt. \tag{4}$$

This is a finite integral of the smooth profile on the compact rectangle, using positive product arclength measure $ds\, dt = ds\, d\delta$.

2.2 A conformal boundary coordinate: condition (C2)

For a metric inducing h, a boundary unit vector has the form $\cos\theta\, T + \sin\theta\, \nu$. The full boundary torus is the abstract space $S = (\mathbb{R}/L\mathbb{Z}) \times (\mathbb{R}/2\pi\mathbb{Z})$. Define a piecewise map on it by

$$\alpha_A(s,\theta) = \begin{cases} (t, 2\pi - \arccos q), & 0 < \theta < \pi, \quad A(s, \cos\theta) = (t, q), \\ (u, \arccos p), & \pi < \theta < 2\pi, \quad A^{-1}(s, \cos\theta) = (u, p), \\ (s, \theta), & \theta = 0, \pi. \end{cases} \tag{5}$$

Require that α_A be smooth on S. Its square is the identity by construction, so it is then a smooth diffeomorphism. For $U \in C^\infty(S; \mathbb{C})$ use the coefficients

$$U_k(s) = \frac{1}{2\pi} \int_0^{2\pi} U(s,\theta) e^{-ik\theta}\, d\theta.$$

Let $\mathcal{B}_A$ consist of the functions U_0 obtained from such U with

$$U \circ \alpha_A = U, \qquad U(s, \theta + \pi) = U(s,\theta), \qquad U_k = 0 \quad (k < 0). \tag{6}$$

This definition uses only functions on the boundary torus.

Set $\omega = e^{2\pi i/3}$. Require that there be exactly one smooth orientation-preserving circle diffeomorphism

$$\phi : \mathbb{R}/L\mathbb{Z} \longrightarrow \mathbb{S}^1, \qquad \phi \in \mathcal{B}_A, \qquad \phi(s_j) = \omega^{j-1} \quad (j = 1, 2, 3). \tag{7}$$

Here membership in $\mathcal{B}_A$ treats ϕ as a complex-valued function. On the unit circle define the positive boundary density b and action D_ϕ by

$$b(\phi(s)) = \frac{1}{|\phi_s(s)|}, \qquad D_\phi(a,c) = D(\phi^{-1}(a), \phi^{-1}(c)) \quad (a \neq c). \tag{8}$$

The angular derivative of an orientation-preserving diffeomorphism is positive, so the denominator never vanishes.

2.3 The convex construction: condition (C3)

All further constructions take place in the fixed closed Euclidean unit disk $\overline{\mathbb{D}}$. Smooth functions on this disk mean restrictions of smooth functions on a neighborhood. Let

$$H = L^2(\mathbb{D}, dx\, dy; \mathbb{R}), \qquad \|u\|_H^2 = \int_{\mathbb{D}} u^2\, dx\, dy,$$

with equality of functions understood almost everywhere. Let $\mathcal{F}$ be the set of $\sigma \in C^\infty(\overline{\mathbb{D}}; \mathbb{R})$, strictly positive on $\overline{\mathbb{D}}$, such that

$$\int_0^1 \sigma(\gamma(r)) |\gamma'(r)|\, dr \geq D_\phi(a,c) \tag{9}$$

for every pair $a \neq c$ on $\mathbb{S}^1$ and every piecewise C^1 curve $\gamma : [0,1] \to \overline{\mathbb{D}}$ joining them. Curves along the boundary, self-intersections, and repeated segments are included. No boundary-value, curvature, or simplicity requirement is imposed on σ in this definition. Let $\mathcal{K}$ be the strong H-closure of $\mathcal{F}$.

Lemma 4.1 shows that $\mathcal{F}$ is nonempty and convex for every map passing (C1)–(C2), and that $\mathcal{K}$ has a unique element ρ_D of smallest H norm. Thus ρ_D is defined before any realizability test on it. Require that this one L^2 class have a smooth representative ρ that is strictly positive on $\overline{\mathbb{D}}$, and require

$$\rho|_{\mathbb{S}^1} = b. \tag{10}$$

A continuous representative, if it exists, is unique, including its boundary values by continuity. Curve integrals are never evaluated on an arbitrary representative of an L^2 class.

2.4 Fixed differential equations: condition (C4)

If (C3) passes, put $\lambda = \log \rho$, $v(\beta) = (\cos\beta, \sin\beta)$, $J(x,y) = (-y,x)$ and $K_\rho = -\rho^{-2}\Delta\lambda$. Require

$$1 + \partial_r \lambda > 0 \quad \text{on } \mathbb{S}^1, \tag{11}$$

where ∂_r is the outward radial derivative. For every interior initial point and every $\beta \in \mathbb{R}/2\pi\mathbb{Z}$, and for every boundary initial point with $z \cdot v(\beta) < 0$, solve the fixed equations

$$z' = \frac{v(\beta)}{\rho(z)}, \qquad \beta' = \frac{\nabla\lambda(z) \cdot Jv(\beta)}{\rho(z)}, \qquad j'' + K_\rho(z) j = 0, \quad j(0) = 0, \quad j'(0) = 1. \tag{12}$$

Require a finite first exit time $\tau > 0$, with $|z(t)| < 1$ for $0 < t < \tau$, and require

$$j(t) > 0 \quad (0 < t \leq \tau). \tag{13}$$

For an inward boundary state, first exit means first return after entering the disk. For an interior state it means its first subsequent encounter with the boundary. The equations are considered only up to this encounter; a smooth extension of ρ outside the disk does not change their solution before it. Boundary tangent states have zero flight time and are excluded.

2.5 Area saturation and the theorem

The final condition (C5) is the exact scalar equality

$$\|\rho_D\|_H^2 = \mathcal{V}(D). \tag{14}$$

Neither side is an approximate or numerical quantity.

Theorem 2.1. *For every smooth compact disk M, every smooth positive boundary metric h, and every map $A : B_h^*\partial M \to B_h^*\partial M$, conditions (C1)–(C5) are necessary and sufficient for A to be the scattering map of a smooth simple Riemannian metric on M inducing h. Their truth is independent of the initial orientation, arclength origin, and positively ordered boundary triple.*

In these conditions, L is the boundary length, D is the primitive (1), ϕ is the uniquely normalized boundary map (7), b is the prescribed circle density (8), and ρ_D is the unique least-norm element of the strong closure defined by (9). All are determined by A, h and the fixed auxiliary choices. In particular, the universal construction of ρ_D is distinct from the conclusion that a realizing metric exists.

Example 2.2. For the Euclidean disk of radius $R > 0$, with $L = 2\pi R$ and marks $0, L/3, 2L/3$, the data are

$$A(s,p) = (s + 2R\arccos p, p), \qquad d(s,\delta) = 2R\sin\frac{\delta}{2R}, \qquad D_{st} = \frac{1}{2R}\sin\frac{\delta}{2R}.$$

Thus $\mathcal{V}(D) = \pi R^2$. The full lift is $\alpha(s,\theta) = (s + 2R\theta, -\theta)$ modulo the two periods. The function $U = e^{is/R}(1 + e^{2i\theta})$ is even, invariant under this lift, and has modes $0, 2$. Its average is $\phi(s) = e^{is/R}$. Lemma 3.3 gives its required uniqueness, and Proposition 4.3 identifies $\rho_D = R$. Here $b = R$, $K_\rho = 0$ and $j(t) = t$. Every permitted ray has finite exit, the expression in (11) is 1, and all five conditions hold. This is an exact calculation of the signs and area factor.

3 Boundary geometry and conformal coordinates

We first consider data from an actual smooth simple metric g inducing h. Let $J_g : TM \to TM$ be the positive g-orthogonal rotation by $\pi/2$; in oriented conformal coordinates it agrees with the Euclidean rotation J in (C4). We use the equivalence between simplicity and strict convexity, nontrapping, and absence of conjugate points on a disk [4, Theorem 3.8.2]. Its connecting geodesics minimize length [4, Proposition 3.8.4].

3.1 The action, twist and area

Lemma 3.1. *For actual simple-metric scattering data, (C1) holds and the primitive D is the boundary distance D_g. The full lift (5) is smooth. Moreover,*

$$\mathrm{Area}_g(M) = \mathcal{V}(D_g). \tag{15}$$

Proof. A geodesic joining distinct boundary points enters and exits strictly transversely. Indeed a boundary defining function positive in the interior has negative second derivative along a tangent geodesic, by strict convexity, so that such a geodesic lies outside near the encounter. The minimizing property also rules out a positive-length return to the starting point.

Let a connecting geodesic vary smoothly with its endpoints. For its constant-speed parametrization $c(r,\varepsilon)$, $0 \le r \le 1$, write $W = \partial_r c$, $Y = \partial_\varepsilon c$ and length $\ell(\varepsilon)$. Metric compatibility, torsion-freeness, and integration by parts give

$$\ell' = \ell^{-1}\int_0^1 g(\nabla_r Y, W)\,dr = g(W/\ell, Y)\big|_{r=1} - g(W/\ell, Y)\big|_{r=0}.$$

Thus $(D_g)_s = -p$ and $(D_g)_t = q$, with the outgoing covector retained.

To calculate the sign of the mixed derivative, fix the initial boundary point and vary the inward angle $0 < \theta < \pi$. The normal Jacobi field is $j(t)J_g\dot c(t)$, where $j(0) = 0$, $j'(0) = 1$ and $j'' + K_g(c(t))j = 0$; see [4, Lemmas 3.7.2, 3.7.6 and 3.7.7]. It is positive until exit, since there are no conjugate points. Write the exit vector as $qT - r\nu$, $r = \sqrt{1-q^2} > 0$. Differentiating the exit event and pairing with $J_g\dot c(\tau)$ yields

$$T\,t_\theta = j(\tau)J_g\dot c(\tau) + \dot c(\tau)\tau_\theta, \qquad r t_\theta = j(\tau).$$

Since $p = \cos\theta$, this gives

$$t_p = -\frac{j(\tau)}{r\sin\theta}, \qquad (D_g)_{st} = -p_t = \frac{r\sin\theta}{j(\tau)} > 0. \tag{16}$$

Unique connecting geodesics give a bijective endpoint map, and this nonzero derivative gives its smooth inverse. The smooth full scattering involution, including the tangent states, is [4, Lemma 3.3.5]; in the (T, ν) frame it is exactly (5). Its incoming restriction gives a smooth diffeomorphism on Z.

The diagonal behavior is checked in endpoint coordinates, without requiring smoothness in p at ± 1. Extend g to a closed surface and to a slightly larger simple neighborhood, as in [4, Lemma 3.1.8 and Proposition 3.8.7]. The exponential inverse $w(s, \delta)$ of the nearby endpoints $x(s), x(s+\delta)$ is smooth with $w(s, 0) = 0$ and $w_\delta(s, 0) = T$. The identity

$$\frac{w(s,\delta)}{\delta} = \int_0^1 w_\delta(s, r\delta)\, dr$$

extends this quotient smoothly and nonvanishingly to $\delta = 0$. For small positive δ the connecting geodesic lies in M and

$$D_g(s, s+\delta) = \delta \sqrt{g_{x(s)}(w(s,\delta)/\delta, w(s,\delta)/\delta)}.$$

It has the required smooth one-sided extension with derivative 1. Symmetry gives the end at $\delta = L$ with derivative -1. Away from the diagonal, smoothness is part of simplicity. First variation and the zero diagonal value identify (1) with D_g, giving all of (C1).

The incoming flux measure is

$$d\mu = \sin\theta\, ds\, d\theta = \, ds\, |dp|.$$

Changing from p to the exit point with $p = -(D_g)_s$ and (16) gives

$$\int_{\partial_+ SM} \tau\, d\mu = \int_{\mathcal{C}} D_g (D_g)_{st}\, ds\, dt.$$

Santaló's formula [4, Proposition 3.6.6] says, for a continuous base function f,

$$\int_{\partial_+ SM} \int_0^\tau f(c(t))\, dt\, d\mu = 2\pi \int_M f\, dV_g. \tag{17}$$

The factor 2π is the mass of the unit tangent circle in Liouville measure $dV_g\, d\theta$ [4, Section 3.5.2]. Taking $f = 1$ proves (15). □

3.2 Holomorphic traces determined by the boundary lift

Let X denote the geodesic vector field on SM, and V the positive fiber rotation. Put $X_\perp = [X, V]$ and $\eta_\pm = (X \pm iX_\perp)/2$. A fiber mode u_k satisfies $-iVu_k = ku_k$; write $\Omega_k = \{u \in C^\infty(SM; \mathbb{C}) : -iVu = ku\}$. Smooth fiber Fourier series and their projections converge in C^∞, and

$$X = \eta_+ + \eta_-, \qquad (Xw)_k = \eta_+ w_{k-1} + \eta_- w_{k+1}, \tag{18}$$

where η_+ raises and η_- lowers the index [4, Lemmas 6.1.3 and 6.1.5]. On base functions, $\eta_- H = 0$ is exactly holomorphicity in the oriented complex structure [4, Lemma 6.1.21]. Vanishing of a mode and the zeroth average are independent of an orthonormal frame rotation. In particular the boundary tangent frame need not extend to the interior.

We use three transport facts with their smooth domains. On a simple surface, a smooth function on ∂SM invariant under the full scattering involution extends uniquely to a smooth w on SM with $Xw = 0$ [4, Theorem 5.1.1]. On such a surface I_0^* is surjective onto $C^\infty(M)$; hence every smooth complex base function H is the zeroth Fourier coefficient of a smooth invariant function

[4, Theorems 8.2.1 and 8.2.2]. Here the source's $\ell_0^* w = 2\pi w_0$ is the unnormalized fiber integral, so the source theorem is applied to $2\pi H$. Finally, on a compact nontrapping surface with strictly convex boundary and surjective I_0^*, the ray transform is injective on each single fiber mode: if u is smooth, $u|_{\partial SM} = 0$, and Xu has one mode, then $u = 0$ [4, Theorem 10.2.3]. Indeed the integral of Xu along every geodesic is zero; injectivity gives $Xu = 0$ and the zero boundary value then gives $u = 0$. Real statements extend to complex functions by applying them to real and imaginary parts in the surjectivity step.

Lemma 3.2. *For actual simple-metric data, $\mathcal{B}_A$ is exactly the set of boundary restrictions of smooth holomorphic base functions on (M, g).*

Proof. Let H be such a holomorphic function. Choose a smooth invariant w with $w_0 = H$ by the preceding surjectivity result, and replace w by $[w(x, v) + w(x, -v)]/2$. It remains invariant, since direction reversal changes the sign of X, and its zeroth average remains H. All its odd modes vanish. Put $W = \sum_{k\geq 0} w_k$. The projection is smooth through the boundary. In (18), every positive mode of XW vanishes by $Xw = 0$, its zeroth mode vanishes by evenness, and its only possible negative mode is $\eta_- w_0 = 0$. Thus $XW = 0$. Its boundary trace U is even, has no negative modes, and is invariant under the full lift. Its average is the trace of H.

Conversely let U satisfy (6). Its invariant extension w is smooth by the transport regularity statement. Uniqueness follows by tracing an interior orbit to the incoming boundary. The reversed $w(x, -v)$ has the same full boundary trace, by the evenness of U, so uniqueness makes w even. Set $w^- = \sum_{k<0} w_k$. It is smooth with zero full boundary trace. The mode equations and evenness give

$$Xw^- = \eta_+ w_{-2} \in \Omega_{-1}.$$

The single-mode injectivity statement applies with $m = -1$ and yields $w^- = 0$. The mode -1 equation for Xw is now $\eta_- w_0 = 0$. Hence w_0 is holomorphic and its boundary trace is U_0. This proves the reverse inclusion. □

3.3 Three-point normalization

Lemma 3.3. *For actual simple-metric data, (C2) holds. Its selected ϕ is the boundary restriction of the unique holomorphic disk diffeomorphism with the prescribed three-point normalization. In this coordinate the metric is $\rho^2 |dz|^2$ with ρ smooth and positive on $\overline{\mathbb{D}}$ and $\rho|_{\mathbb{S}^1} = b$.*

Proof. Smooth uniformization [4, Theorem 3.4.15] supplies a holomorphic map $\Phi_0 : M \to \overline{\mathbb{D}}$ that is a smooth diffeomorphism of the closed disks. Its boundary trace belongs to $\mathcal{B}_A$ by Lemma 3.2.

We first show that every smooth holomorphic H whose boundary trace is an orientation-preserving circle diffeomorphism is itself a disk diffeomorphism. In the coordinate Φ_0, write $F = H \circ \Phi_0^{-1}$. It is holomorphic on $\mathbb{D}$, smooth on its closure and nonconstant. For a fixed w with $|w| \neq 1$, the functions $F - w$ are nonzero near the boundary. The argument principle on circles $|z| = r < 1$, close to 1, gives the zero count

$$n_r(w) = \frac{1}{2\pi i} \int_{|z|=r} \frac{F'(z)}{F(z) - w}\, dz.$$

As $r \uparrow 1$ this tends to the corresponding boundary integral. Writing $F(e^{i\theta}) = e^{ia(\theta)}$, where $a' > 0$ and $a(\theta + 2\pi) = a(\theta) + 2\pi$, and changing variable to a shows that the limit is 1 for $|w| < 1$ and 0 for $|w| > 1$. For example these values follow by expanding the denominator in its uniformly convergent geometric series on the boundary circle. The integer $n_r(w)$ therefore equals its limit for all r sufficiently close to 1. In the first case there are no zeros in a thin outer annulus, so there is

exactly one preimage in $\mathbb{D}$, with multiplicity one. In the second case there is none: any preimage would lie inside one of the circles with zero count.

An interior preimage of $w_0 \in \mathbb{S}^1$ is also impossible. A small circle around such a point has a positive argument-principle count for $F - w_0$. That integer remains unchanged for w near w_0, by continuity and nonvanishing on the small contour. Taking $|w| > 1$ contradicts the zero count just proved. Thus F maps $\mathbb{D}$ bijectively onto $\mathbb{D}$ and its derivative never vanishes there. At the boundary its tangential derivative is nonzero. The Cauchy–Riemann equations hold there by continuity, so its real Jacobian is nonzero as well. A smooth extension and the inverse function theorem give smooth local inverses through the boundary. All uses of the argument principle were on interior contours; no analytic continuation through $\mathbb{S}^1$ was assumed.

For completeness, the three-point normalization can be made explicitly. For a positively ordered triple $\zeta_1, \zeta_2, \zeta_3$ on $\mathbb{S}^1$, set

$$C_{\zeta_3}(z) = i\frac{\zeta_3 + z}{\zeta_3 - z}, \qquad r_j = C_{\zeta_3}(\zeta_j)\ (j = 1, 2), \qquad H_\zeta = \frac{C_{\zeta_3} - r_1}{r_2 - r_1}.$$

The imaginary part of C_{ζ_3} is $(1-|z|^2)/|\zeta_3 - z|^2$, and $r_1 < r_2$ are real by the cyclic order. Thus H_ζ maps the disk to the upper half-plane and the triple to $(0, 1, \infty)$. Applying the same formula to $(1, \omega, \omega^2)$, the composition $H_{\text{roots}}^{-1} H_\zeta$ is the desired disk automorphism. Upper half-plane automorphisms are real fractional linear maps [3, Theorem 1.7], and one fixing $0, 1, \infty$ is the identity. Any competing $\phi \in \mathcal{B}_A$ extends by Lemma 3.2 to a holomorphic H, which is a disk diffeomorphism by the preceding argument. Comparing the two disk maps therefore proves uniqueness.

In the normalized coordinate the pushed metric is $\rho^2|dz|^2$, with smooth positive ρ through the boundary. The boundary arclength identity is

$$\rho(\phi(s))|\phi_s(s)| = 1,$$

which gives exactly (8). Together with Lemma 3.1, this proves (C2). □

4 Variational realization

We now begin with proposed data, without assuming any realizing metric. The first lemma makes the convex construction well defined on every input passing its boundary prerequisites.

4.1 Universal feasibility and projection

Lemma 4.1. *Under (C1)–(C2), $\mathcal{F}$ is nonempty and convex. Its strong H closure $\mathcal{K}$ has a unique element ρ_D of smallest norm. If this class has a continuous representative on $\overline{\mathbb{D}}$, that representative is unique.*

Proof. Since $|D_t| = |Q| < 1$ and the profile vanishes at both ends,

$$D(s, s + \delta) \le \delta, \qquad D(s, s + \delta) \le L - \delta.$$

Thus D is bounded by the shorter h-arclength arc. Put $B = \max_{\mathbb{S}^1} b$. If $a, c \in \mathbb{S}^1$ have shorter angular separation $\alpha \in (0, \pi]$, the corresponding h-shorter arc is no longer than $B\alpha$. Since $\alpha \le (\pi/2)|a - c|$,

$$D_\phi(a, c) \le \frac{\pi}{2} B|a - c|.$$

Every joining piecewise C^1 curve has Euclidean length at least $|a - c|$, so the constant $\sigma = (\pi/2)B$ belongs to $\mathcal{F}$.

Strict positivity and the linear inequalities (9) persist under convex combinations. Hence $\mathcal{F}$, and also its closure, are convex. Let $m = \inf_{u\in\mathcal{K}} \|u\|_H$. Choose $u_n \in \mathcal{K}$ with $\|u_n\|_H^2 \le m^2 + 1/n$. The midpoint is in $\mathcal{K}$, and the parallelogram identity gives

$$\|u_n - u_k\|_H^2 = 2\|u_n\|_H^2 + 2\|u_k\|_H^2 - 4\|(u_n + u_k)/2\|_H^2 \le 2/n + 2/k.$$

Completeness of H and closedness of $\mathcal{K}$ produce a limit of norm m. Applying the same identity to two minima proves uniqueness. If continuous representatives differed at an interior point, their difference would be bounded away from zero on a ball of positive area, contradicting equality almost everywhere. Interior limits then identify their boundary values. □

4.2 The ODE tests and necessity

Lemma 4.2. *For a smooth positive ρ on $\overline{\mathbb{D}}$, condition (C4) is equivalent to simplicity of $g_\rho = \rho^2|dz|^2$.*

Proof. The conformal geodesic equation has the first two components in (12) [4, Lemma 3.5.6]. Its speed has g_ρ norm one. More explicitly the Christoffel symbols are

$$\Gamma_{ij}^k = \delta_i^k \lambda_j + \delta_j^k \lambda_i - \delta_{ij}\lambda_k.$$

In the curvature expression the quadratic first-derivative terms cancel, giving $K_{g_\rho} = -\rho^{-2}(\lambda_{xx} + \lambda_{yy})$. On the unit circle the unit tangent and inward normal are respectively $\rho^{-1}(-\sin\psi, \cos\psi)$ and $-\rho^{-1}(\cos\psi, \sin\psi)$. Substitution gives

$$g_\rho(\nabla_T T, \nu) = \rho^{-1}(1 + \partial_r \lambda).$$

Thus (11) is exactly strict boundary convexity. The third equation in (12) is the normal Jacobi equation [4, Lemmas 3.7.6 and 3.7.7].

Finite exit for all interior and inward boundary initial states gives forward nontrapping; reversing the initial directions gives backward nontrapping. A normal Jacobi field vanishing at its initial point is a multiple of j. Its positivity through exit prevents conjugate pairs. The tangential Jacobi component has the form $a + bt$ and cannot vanish at two distinct times unless it is zero. Every subsegment begins at a state covered by (C4), so this includes conjugate pairs starting in the interior and pairs involving endpoints. The equivalence in [4, Theorem 3.8.2] now gives simplicity. Conversely simplicity gives finite exit and no zero of j after its initial positive slope, as well as strict convexity, so all the ODE tests hold. □

Proposition 4.3. *Every smooth simple Riemannian metric inducing h produces data satisfying (C1)–(C5). Its normalized conformal density is exactly ρ_D.*

Proof. Lemmas 3.1 and 3.3 give (C1)–(C2) and a normalized smooth positive density ρ with boundary value b. Every curve has g_ρ length at least the boundary distance, so $\rho \in \mathcal{F}$. For any $\sigma \in \mathcal{F}$ and any incoming g_ρ-unit-speed geodesic, (9) gives

$$I_g(\sigma/\rho) := \int_0^\tau (\sigma/\rho)(c(t))\, dt \ge \tau.$$

The endpoints are distinct. Integrate over the incoming boundary and apply (17) to the continuous base functions σ/ρ and 1. In Euclidean coordinates this yields

$$2\pi \int_{\mathbb{D}} \sigma\rho \, dx\, dy \ge 2\pi \int_{\mathbb{D}} \rho^2 \, dx\, dy.$$

The pairing with ρ is continuous in H, so $\langle u, \rho\rangle_H \geq \|\rho\|_H^2$ for every $u \in \mathcal{K}$. Consequently

$$\|u\|_H^2 = \|\rho\|_H^2 + \|u-\rho\|_H^2 + 2\langle u-\rho, \rho\rangle_H \geq \|\rho\|_H^2 + \|u-\rho\|_H^2.$$

Thus the true density is the unique least-norm element ρ_D. It satisfies (C3), and Lemma 4.2 gives (C4). Finally (15) gives $\|\rho_D\|_H^2 = \mathrm{Area}_g(M) = \mathcal{V}(D)$, which is (C5). □

4.3 Passing length inequalities through the closure

Lemma 4.4. *Assume (C1)–(C4), and use ϕ to express the boundary distance of the selected simple metric g_ρ in the original arclength coordinates. Then*

$$D_{g_\rho}(s,t) \geq D(s,t) \quad (s \neq t). \tag{19}$$

Proof. The selected density has the correct boundary metric since $\rho(\phi(s))|\phi_s(s)| = 1$. Choose $\sigma_n \in \mathcal{F}$ with $\sigma_n \to \rho$ strongly in H, and put $f_n = (\sigma_n - \rho)/\rho$. All these functions are smooth. Santaló's formula and Cauchy–Schwarz give

$$\begin{aligned}\int_{\partial_+ S\overline{\mathbb{D}}} |I_g f_n|\, d\mu &\leq \int_{\partial_+ S\overline{\mathbb{D}}} I_g|f_n|\, d\mu \\ &= 2\pi \int_{\mathbb{D}} |\sigma_n - \rho|\rho\, dx\, dy \leq 2\pi \|\sigma_n - \rho\|_H \|\rho\|_H \longrightarrow 0.\end{aligned} \tag{20}$$

For each of its geodesics, the curve inequality gives $\tau + I_g f_n \geq D(s,t)$. Therefore

$$(D(s,t) - \tau)_+ \leq |I_g f_n|.$$

The integral of this nonnegative left side is zero by (20). It is continuous on the open incoming bundle, where flux has strictly positive density. Positivity at one state would give a neighborhood of positive integral; hence it vanishes everywhere. The endpoint diffeomorphism of Lemma 3.1 then gives (19) for every distinct boundary pair. Only smooth functions were integrated along individual curves. No trace on a curve or ray transform of a general L^2 class was used. □

4.4 Strict area monotonicity and sufficiency

Lemma 4.5. *Let D_0, D_1 have smooth zero-ended profiles as in (2), and suppose $(D_i)_{st} > 0$ on $\mathcal{C}$. If $D_1 \geq D_0$ and $\mathcal{V}(D_1) = \mathcal{V}(D_0)$, then $D_1 = D_0$.*

Proof. On the compact profile rectangle put $\mathcal{L} = \partial_s\partial_\delta - \partial_\delta^2$. For smooth periodic zero-ended profiles u, v, two integrations by parts give

$$\int uv_{s\delta} = -\int u_\delta v_s = \int u_{s\delta} v, \qquad -\int uv_{\delta\delta} = \int u_\delta v_\delta = -\int u_{\delta\delta} v.$$

Terms at $s = 0, L$ cancel by periodicity, while the first and second δ boundary terms vanish because $u = 0$ and $v = 0$, respectively. Thus $\int u\mathcal{L}v = \int v\mathcal{L}u$. Expanding the quadratic functional (4) gives

$$\mathcal{V}(D_1) - \mathcal{V}(D_0) = \frac{1}{2\pi}\int_{\mathcal{C}} (D_1 - D_0)((D_1)_{st} + (D_0)_{st})\, ds\, dt. \tag{21}$$

If $D_1 - D_0$ is nonzero and nonnegative, continuity makes it positive on an open set. The mixed derivatives are positive there, so the right side is strictly positive, a contradiction. All integrations by parts occur on the smooth compact rectangle; no diagonal-strip limit is needed. □

Proof of Theorem 2.1. Necessity is Proposition 4.3. For sufficiency assume all five conditions. Lemma 4.1 defines the selected density, (C3) supplies its smooth positive representative, and Lemma 4.2 makes g_ρ simple. Lemma 4.4 gives $D_{g_\rho} \geq D$. The two actions have smooth zero-ended profiles and positive mixed derivative, by (C1) and Lemma 3.1. Their action areas agree, since

$$\mathcal{V}(D_{g_\rho}) = \mathrm{Area}_{g_\rho}(\overline{\mathbb{D}}) = \|\rho_D\|_H^2 = \mathcal{V}(D).$$

Lemma 4.5 forces $D_{g_\rho} = D$ pointwise.

For $(s,p) \in Z$ let $A(s,p) = (t,q)$. The unique geodesic from s to t for the selected metric has initial tangential covector $-(D_{g_\rho})_s = -D_s = P(s,t) = p$ and exit covector $(D_{g_\rho})_t = D_t = Q(s,t) = q$. Thus its scattering is exactly A on the whole open cylinder.

To realize this metric on the fixed smooth disk M, choose any orientation-preserving smooth disk chart $M \to \overline{\mathbb{D}}$. In this chart the desired boundary map is $e^{i\theta} \mapsto e^{ia(\theta)}$ with $a' > 0$ and $a(\theta + 2\pi) = a(\theta) + 2\pi$. Choose a smooth radial cutoff $0 \leq \chi \leq 1$, zero near the origin and one near the boundary. The map

$$(r,\theta) \longmapsto (r, \theta + \chi(r)(a(\theta) - \theta))$$

is well defined modulo 2π and has positive angular derivative $1 - \chi + \chi a'$. It is the identity near the center and a smooth bijection on each circle, with smooth inverse. Composing with the chart gives a smooth disk diffeomorphism Ψ with boundary value ϕ. The pullback $\Psi^*(\rho^2|dz|^2)$ is a smooth simple Riemannian metric on M. It induces h by (10) and (8), and its scattering is the prescribed A.

Every initial choice was arbitrary in both directions of the proof. Passing the tests for one choice gives a realizing metric; necessity then makes its data pass for every other choice. This proves the claimed independence of the classification. For proposed data without a realization, no algebra or fractional-linear closure property of $\mathcal{B}_A$ has been assumed. □

AI-use statement. An OpenAI Codex agent assisted in developing the argument, checking cited source statements, and preparing this manuscript. The submitted result is an analytic proof claim. AI-assisted checks and successful typesetting do not establish independent mathematical verification. Independent review of the complete argument is requested.

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